

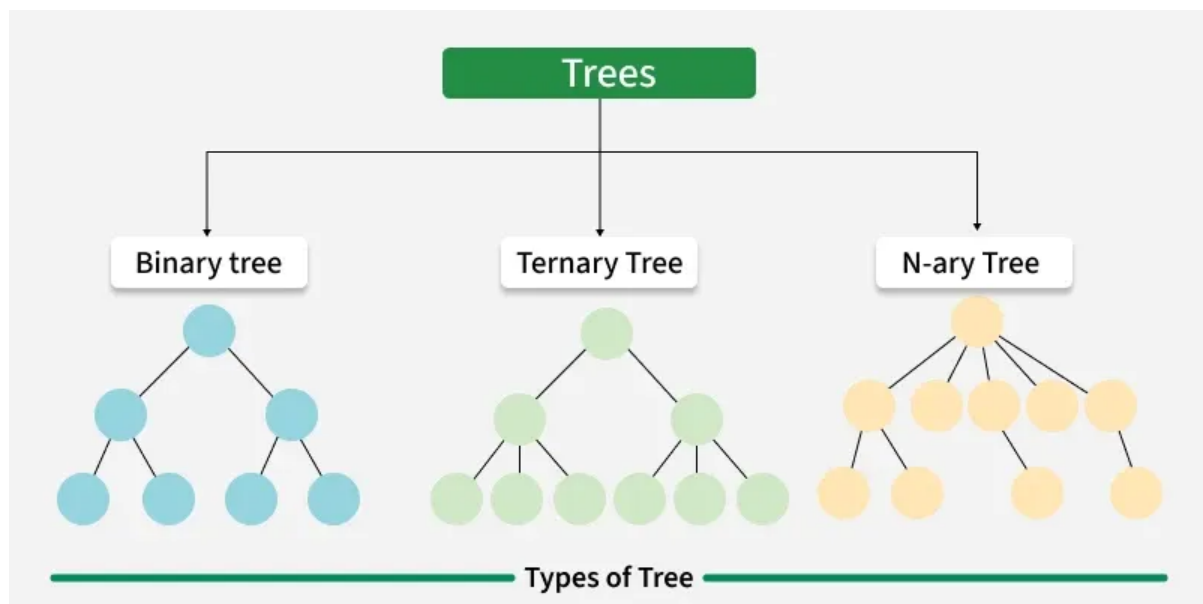
Types of Trees in Data Structures

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A [tree](#) in data structures is a hierarchical data structure that consists of nodes connected by edges. It is used to represent relationships between elements, where each node holds data and is connected to other nodes in a parent-child relationship.

Types of Trees



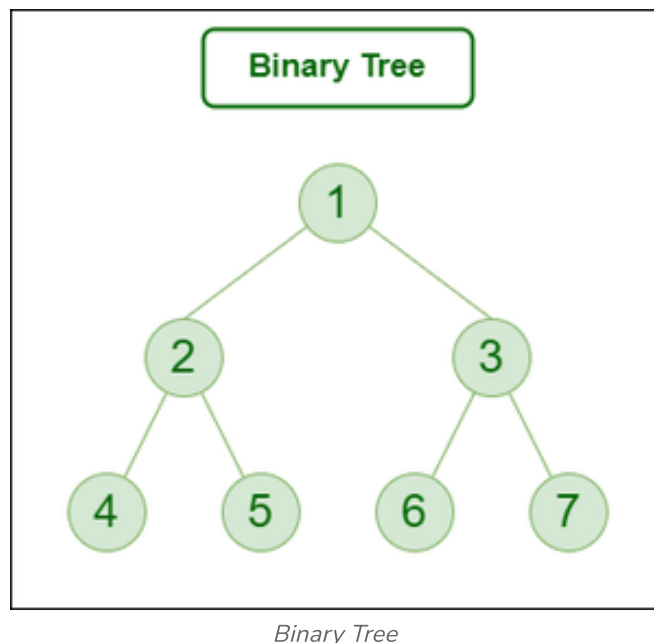
Tree

The main types of trees in data structure are:

1. Binary Tree

*A binary tree is a tree data structure where each node has at most **two children**. These two children are usually referred to as the **left child** and **right child**. It is widely used in applications such as binary search trees and heaps.*

Example: Consider the tree below. Since each node of this tree has only 2 children, it can be said that this tree is a [Binary Tree](#)

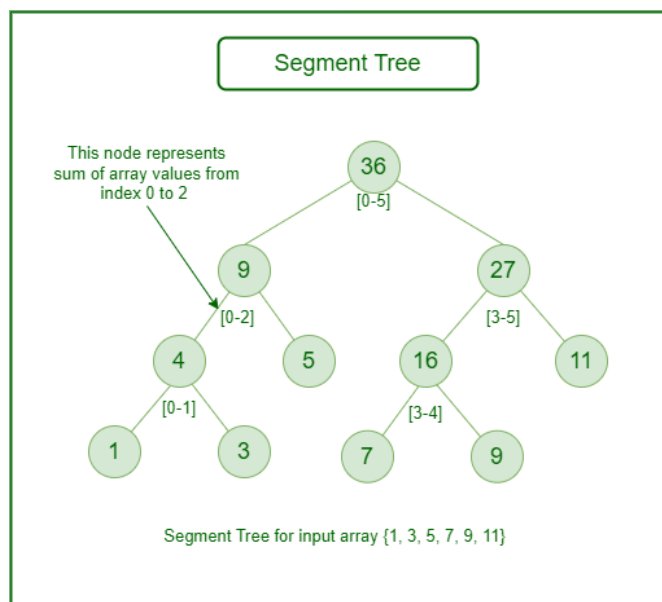


Examples / Types of Binary Tree:

- **Complete Binary Tree:** A binary tree in which every level, except possibly the last, is completely filled, and all nodes are as far left as possible. For example, a [heap](#) is a complete binary tree that satisfies the heap property (either max-heap or min-heap).
- **Full Binary Tree:** A binary tree where every node has either 0 or 2 children.
- **Degenerate Binary Tree (or Pathological Tree):** A tree in which each parent node has only one child. This essentially forms a linked list, which leads to inefficient operations.
- **Binary Search Tree (BST) and its Variations:** A [BST](#) is a binary tree where each node has at most two children, and for each node, the left child's value is smaller than the node's value, and the right child's value is greater. An [AVL Tree](#) is a self-balancing BST where the heights of the two child subtrees of any node differ by no more than one. [Red Black Tree](#) and [Splay Tree](#) are more example variations of BST.
- **Binary Indexed Tree ([Fenwick Tree](#)):** A data structure that uses a binary tree to efficiently compute and update prefix sums in an array.
- **Perfect Binary Tree:** A binary tree where all internal nodes have two children and all leaf nodes are at the same level.
- **Balanced Binary Tree:** A binary tree where the difference in heights between the left and right subtrees of any node is minimal (often defined as at most 1).

Examples of Balanced Binary Tree are [AVL Tree](#), [Red Black Tree](#) and [Splay Tree](#)

- **Segment Tree:** A [segment tree](#) is a binary tree used for storing intervals or segments. It allows efficient querying and updating of ranges of values, making it particularly useful for problems that involve finding the sum, minimum, maximum, or other operations over a range of elements in an array.

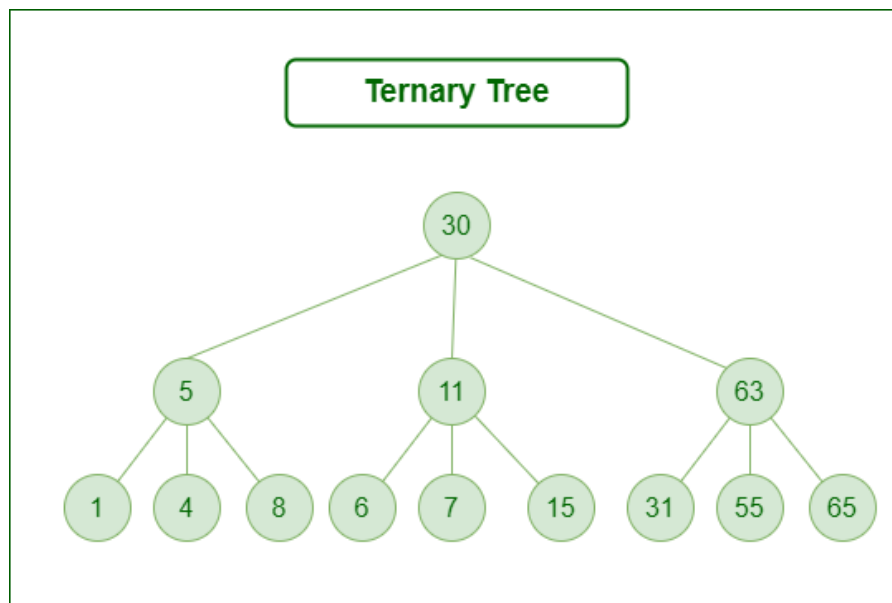


To know more about the types of the binary trees, please refer to this detailed article: [Types of Binary Tree](#).

2. Ternary Tree

A Ternary Tree is a tree data structure in which each node has at most three child nodes, usually distinguished as “left”, “mid” and “right”.

Example: Consider the tree below. Since each node of this tree has only 3 children, it can be said that this tree is a [Ternary Tree](#).



Examples of Ternary Tree:

- **Ternary Search Tree:** A [ternary search tree](#) is a special trie data structure where the child nodes of a standard trie are ordered as a binary search tree.
- **Ternary Heap:** A type of heap where each node can have up to three children, though less common than binary heaps.

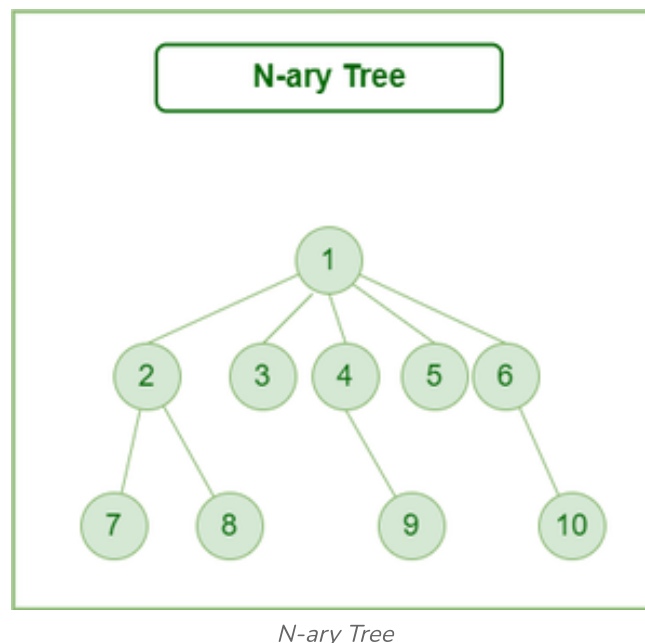
3. N-ary Tree (Generic Tree)

[Generic trees](#) are a collection of nodes where each node is a data structure that consists of records and a list of references to its children(duplicate references are not allowed). Unlike the linked list, each node stores the address of multiple nodes.

Every node stores the addresses of its children and the very first node's address will be stored in a separate pointer called root.

1. Many children at every node.
2. The number of nodes for each node is not known in advance.

Example:



Examples of N-ary Trees:

- **B-tree:** A [self-balancing search tree](#) where nodes can have multiple children, usually used for indexing large databases.
- **B+ Tree:** A [B+ tree](#) is a variation of the B-tree that only stores data in the leaf nodes, making range queries more efficient.
- **Trie (Prefix Tree):** A tree where each node represents a character, and paths from the root to leaves represent strings. It can have a variable number of children for each node, making it an N-ary tree.

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Binary Tree



Ternary Tree



